



Control design for aeronautical vehicles

Daniel Alazard

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This document contains:

- a copy of the slides used during this course,
- the statement of exercices and Matlab Labwork.

EMPTY PAGE

Control Design for aerospace vehicles

D. Alazard¹



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Institut Supérieur de l'Aéronautique et de l'Espace

Outline I

- 1 **State-feedback for eigen-structure assignment**
 - Single input case
 - Multi input case
- 2 **Linear Quadratic (LQ) state feedback design**
- 3 **Norms for linear systems**
- 4 **H_∞ requirements for servo-loop systems**
- 5 **H_∞ control**
 - General results
 - Example on the interest of structured-controllers
- 6 **H_2 control**

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State feedback design - Single input case.

Principle: Considering a n -th order linear system given by its state space representation and assuming that all the state variables are measured:

$$\begin{cases} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}u, & \mathbf{x} \in \mathbb{R}^n, u \in \mathbb{R} \\ \mathbf{y} &= \mathbf{x} \end{cases}$$

the objective is to compute a state feedback gain $\mathbf{K}_{1 \times n}$ such that the control law $u = e - \mathbf{K}\mathbf{x}$ assigns the n closed-loop eigenvalues to prescribed values $\lambda_i, i = 1, \dots, n$.

λ_i are the tuning parameters, chosen such that the closed loop system is:

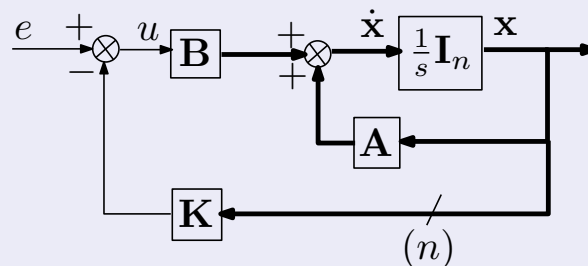
- stable: $\text{Re}(\lambda_i) < 0 \forall i$,
- correctly damped: $-\text{Re}(\lambda_i)/\sqrt{\lambda_i \bar{\lambda}_i} \approx 0.7$,
- fast enough (for the dominant dynamics): $\text{Re}(\lambda_i) < -1/\tau$,
- not too far from the open-loop dynamics: $\text{Spec}(\mathbf{A})$,
- ...

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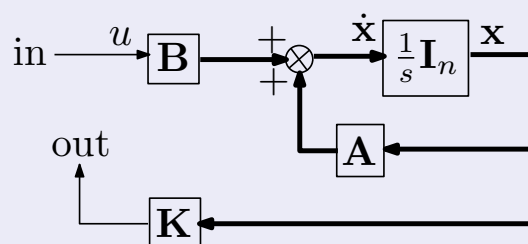
State feedback design - Single input case.

Closed-loop system representation: $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{BK})\mathbf{x} + \mathbf{B}e$



Open-loop transfer function $L(s)$ for a state feedback:

$$L(s) = \mathbf{K}(s\mathbf{I}_n - \mathbf{A})^{-1}\mathbf{B}$$



Stability margins (in MATLAB): `margin(ss(A,B,K,0)).`

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State feedback - Computation of the gain $\mathbf{K}$.

Assumption: the pair $(\mathbf{A}, \mathbf{B})$ is controllable.

$\Rightarrow$ 2 methods

Method # 1: Direct identification of the closed-loop characteristic polynomial.

Let $\mathbf{K} = [k_0, k_1, \dots, k_{n-1}]$ then

$$\det(s\mathbf{I}_n - \mathbf{A} + \mathbf{BK}) = \prod_{i=1}^n (s - \lambda_i) = s^n + a_{n-1}s^{n-1} + a_{n-2}s^{n-2} + \dots + a_1s + a_0.$$

The identification of polynome coefficients leads to a set of n equations with n unknowns $(k_i, i = 0, \dots, n-1)$ to be solved.

Example:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 10 \\ 0 & 0 & -10 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

Compute a state feedback $\mathbf{K}$ such that the closed-loop dynamics is governed by a second order with a pulsation $\omega (> 0)$ and a damping ratio $\xi (> 0)$.

State feedback - Computation of the gain $\mathbf{K}$.

Method # 2: assumption: λ_i are distinct.

$$\lambda_i \in \text{Spec}(\mathbf{A} - \mathbf{BK}) \Leftrightarrow \exists \mathbf{v}_i (\text{eigenvector}) / (\mathbf{A} - \mathbf{BK})\mathbf{v}_i = \lambda_i \mathbf{v}_i$$

$$\Leftrightarrow \begin{bmatrix} \mathbf{v}_i \\ w_i \end{bmatrix} \in \text{Ker}([\mathbf{A} - \lambda_i \mathbf{I}_n \quad \mathbf{B}]) \quad \text{with} \quad w_i = -\mathbf{K}\mathbf{v}_i. \quad (1)$$

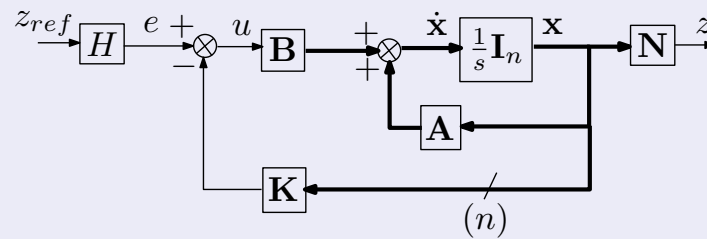
The direction $[\mathbf{v}_i^T \ w_i^T]^T$ (up to a factor) must be computed for each λ_i . Then $\mathbf{K}$ must satisfy: $w_i = -\mathbf{K}\mathbf{v}_i, \forall i$

$$\Rightarrow \mathbf{K} = - \underbrace{[w_1 \ w_2 \ \dots \ w_n]}_{\mathbf{W}} \underbrace{[\mathbf{v}_1 \ \mathbf{v}_2 \ \dots \ \mathbf{v}_n]}_{\mathbf{V}}^{-1} = -\mathbf{WV}^{-1}. \quad (2)$$

Example: solve the previous example (slide # 7) using MATLAB and the function `null`.

State feedback and feedforward

Considering a single output $z = \mathbf{N}\mathbf{x}$ to be “servo-looped” on the reference input z_{ref} using a state-feedback gain $\mathbf{K}$ and a feedforward gain H : $\Rightarrow u = H z_{ref} - \mathbf{K}\mathbf{x}$.



$$\Rightarrow \frac{Z}{Z_{ref}}(s) = \mathbf{N}(s\mathbf{I}_n - \mathbf{A} + \mathbf{BK})^{-1}\mathbf{B}H.$$

H can be computed to impose an unitary DC-gain between z_{ref} and z :

$$H = -(\mathbf{N}(\mathbf{A} - \mathbf{BK})^{-1}\mathbf{B})^{-1}.$$

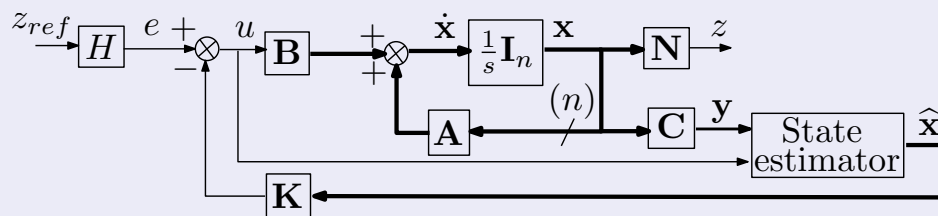
Example: Considering the example of slide # 7, compute H to servo-loop the first state $x(1)$.

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State feedback control: limitations

From a practical point of view, all the state variables are not always measured. Let $\mathbf{y} = \mathbf{C}_{p \times n}\mathbf{x}$ the vector of the p measured outputs. Then, the state feedback gain $\mathbf{K}$ can be implemented using a state estimator:
 $u = H z_{ref} - \mathbf{K}\hat{\mathbf{x}}$.



In the field of Aerospace vehicle control, such a state estimator is also called a **Navigation Filter** (see course on KALMAN filter).

Alternative: one can also use a static output feedback $\mathbf{K}_y \in \mathbb{R}^{1 \times p}$ such that the control law $u = H z_{ref} - \mathbf{K}_y \mathbf{y}$ assigns **only** p closed-loop eigenvalues to prescribed values λ_i , $i = 1, \dots, p$.

Then (using the method # 2 and equation (1) with $w_i = -\mathbf{K}_y \mathbf{C}\mathbf{v}_i$):

$$\mathbf{K}_y = -[w_1 \cdots w_p] (\mathbf{C}[\mathbf{v}_1 \cdots \mathbf{v}_p])^{-1}; H = -(\mathbf{N}(\mathbf{A} - \mathbf{BK}_y\mathbf{C})^{-1}\mathbf{B})^{-1}.$$

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Static output feedback control (K_y)

Example:

$$\begin{aligned}\dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 10 \\ 0 & 0 & -10 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \\ \mathbf{y} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x}\end{aligned}$$

Compute a static output feedback $\mathbf{K}_y$ such that the closed-loop dynamics is governed by a second order with a pulsation $\omega (> 0)$ and a damping ratio $\xi (> 0)$. What condition ω and ξ must satisfy for the closed-loop system to be stable ?

Case of a directfeedthrough: $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$ and $\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}u$.

Considering $\tilde{\mathbf{y}} = \mathbf{y} - \mathbf{D}u$, the previous method provides: $\mathbf{K}_{\tilde{\mathbf{y}}}$ and $\tilde{H}$. Then:

$$\mathbf{K}_y = \mathbf{K}_{\tilde{\mathbf{y}}}(\mathbf{I}_p - \mathbf{D}\mathbf{K}_{\tilde{\mathbf{y}}})^{-1}; \quad H = (1 + \mathbf{K}_y\mathbf{D})\tilde{H}$$

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State feedback design - Multi input case.

Considering that $\mathbf{u} \in \mathbb{R}^m$ ($\Rightarrow \mathbf{B} \in \mathbb{R}^{n \times m}$), then

$\text{Ker} \left(\begin{bmatrix} \mathbf{A} - \lambda_i \mathbf{I}_n & \mathbf{B} \end{bmatrix}_{n \times (n+m)} \right)$ is now a m dimension sub-space (under the controllability assumption).

In addition to assign the eigenvalue λ_i , it is thus possible to assign $m - 1$ components (or combinations of components) of the eigenvector $\mathbf{v}_i$ or of the eigen-input-direction $\mathbf{w}_i = -\mathbf{K}\mathbf{v}_i$ associated to λ_i . That is, till to $m - 1$ constraints $\mathbf{C}_i \mathbf{v}_i + \mathbf{D}_i \mathbf{w}_i = 0$ with $\mathbf{C}_i \in \mathbb{R}^{(m-1) \times n}$ and $\mathbf{D}_i \in \mathbb{R}^{(m-1) \times m}$ can be taken into account, choosing:

$$\begin{bmatrix} \mathbf{v}_i \\ \mathbf{w}_i \end{bmatrix} \in \text{Ker} \left(\begin{bmatrix} \mathbf{A} - \lambda_i \mathbf{I}_n & \mathbf{B} \\ \mathbf{C}_i & \mathbf{D}_i \end{bmatrix} \right).$$

Indeed $\mathbf{K}$ is now a $m \times n$ matrix of free parameters which can be used to assigned the n closed-loop eigenvalues and to handle eigen-vectors (or decouplings) specifications.

Examples of decoupling specifications:

- to cancel the k -th ($1 \leq k \leq n$) component of v_i :

$$\mathbf{C}_i = [\mathbf{0}_{1 \times (k-1)}, 1, \mathbf{0}_{1 \times (n-k)}], \mathbf{D}_i = \mathbf{0}_{1 \times m},$$

- to cancel the effect of the closed-loop eigenvalues λ_i on the output $\mathbf{z} = \mathbf{N}\mathbf{x}$:

$$\mathbf{C}_i = \mathbf{N}, \mathbf{D}_i = \mathbf{0}_{1 \times m},$$

- to cancel the effect of the closed-loop eigenvalues λ_i on k -th ($1 \leq k \leq n$) control signal u_k :

$$\mathbf{C}_i = \mathbf{0}_{1 \times n}, \mathbf{D}_i = [\mathbf{0}_{1 \times (k-1)}, 1, \mathbf{0}_{1 \times (m-k)}].$$

Then, $\mathbf{v}_i$ and $\mathbf{w}_i$ can be computed for each λ_i , $i = 1, \dots, n$ and $\mathbf{K}$ is still given by equation (2).

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LQ synthesis

Let us considering the linear system:

$$\begin{cases} \dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) & \mathbf{x} \in \mathbb{R}^n; \mathbf{u} \in \mathbb{R}^m \\ \mathbf{z}(t) &= \mathbf{N}\mathbf{x}(t) & \mathbf{z} \in \mathbb{R}^q. \end{cases}$$

Assumption: The pair $(\mathbf{A}, \mathbf{B})$ is stabilizable.

Result: The state feedback control which stabilizes the system and minimize the quadratic performance index:

$$J = \int_0^\infty (\mathbf{z}^T(t)\mathbf{Q}\mathbf{z}(t) + \mathbf{u}^T(t)\mathbf{R}\mathbf{u}(t))dt = \int_0^\infty (\mathbf{x}^T(t)\mathbf{Q}_x\mathbf{x}(t) + \mathbf{u}^T(t)\mathbf{R}\mathbf{u}(t))dt$$

with $\mathbf{R} > 0$, $\mathbf{Q} \geq 0$ ($\mathbf{Q}_x = \mathbf{N}^T\mathbf{Q}\mathbf{N}$) reads:

$$\mathbf{u}(t) = -\mathbf{K}_c\mathbf{x}(t) \text{ with } \mathbf{K}_c = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}_c$$

where $\mathbf{P}_c$ is positive (symmetric) solution of algebraic RICCATI equation:

$$\mathbf{P}_c\mathbf{A} + \mathbf{A}^T\mathbf{P}_c - \mathbf{P}_c\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}_c + \mathbf{Q}_x = 0. \quad (3)$$

Then: $J_{\min} = \mathbf{x}_0^T\mathbf{P}_c\mathbf{x}_0$ ($\mathbf{x}_0$: initial state at $t = 0$).

Let $\mathbf{L}_c(s) = \mathbf{K}_c(s\mathbf{I}_n - \mathbf{A})^{-1}\mathbf{B}$ and $\mathbf{G}_{zu}(s) = \mathbf{N}(s\mathbf{I}_n - \mathbf{A})^{-1}\mathbf{B}$, then it can be shown:

$$(\mathbf{I}_m + \mathbf{L}_c^T(-s))\mathbf{R}(\mathbf{I}_m + \mathbf{L}_c(s)) = \mathbf{R} + \mathbf{G}_{zu}^T(-s)\mathbf{Q}\mathbf{G}_{zu}(s) \quad (4)$$

In the single input case ($\dim u = 1$), we get:

$$|1 + L_c(j\omega)| \geq 1 \quad \text{or} \quad |S_u(j\omega)| \leq 1, \forall \omega.$$

Thus stability margins are quite comfortable:

Gain margin $\notin [0.5; +\infty]$, Phase margin $\notin [-60^\circ; +60^\circ]$

Modulus margin ≥ 1 .

This property is lost if the quadratic performance index contains state/control cross terms:

$$J = \int_0^\infty (\mathbf{x}^T \mathbf{Q}_x \mathbf{x} + 2\mathbf{x}^T \mathbf{S} \mathbf{u} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt.$$

LQ synthesis - Riccati equation solver

(3) equation can be also expressed:

$$[\mathbf{P}_c \quad -\mathbf{I}_n] \begin{bmatrix} \mathbf{A} & -\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T \\ -\mathbf{Q}_x & -\mathbf{A}^T \end{bmatrix} \begin{bmatrix} \mathbf{I}_n \\ \mathbf{P}_c \end{bmatrix} = \mathbf{0}.$$

$\mathbf{H} = \begin{bmatrix} \mathbf{A} & -\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T \\ -\mathbf{Q}_x & -\mathbf{A}^T \end{bmatrix}$ is called the **hamiltonian** matrix

associated to the RICCATI equation (3).

Property of $\mathbf{H}$: the $2n$ eigenvalues of $\mathbf{H}$ are:

- the n “stable” eigenvalues of the closed-loop system $\text{spec}(\mathbf{A} - \mathbf{B}\mathbf{K}_c)$,
- the n eigenvalues symmetric to $\text{spec}(\mathbf{A} - \mathbf{B}\mathbf{K}_c)$ w.r.t. the imaginary axis in the complex plane (thus “unstable”).

Proof trick: considering the mapping on $\mathbf{H}$:

$$\tilde{\mathbf{H}} = \mathbf{M}^{-1}\mathbf{H}\mathbf{M} \quad \text{with} \quad \mathbf{M} = \begin{bmatrix} \mathbf{I}_n & \mathbf{0} \\ \mathbf{P}_c & \mathbf{I}_n \end{bmatrix} \quad \text{and} \quad \mathbf{M}^{-1} = \begin{bmatrix} \mathbf{I}_n & \mathbf{0} \\ -\mathbf{P}_c & \mathbf{I}_n \end{bmatrix},$$

$\Rightarrow$ (3) is solved using invariant sub-spaces decomposition:

LQ synthesis - asymptotic properties

General theorem: Considering the linear system:

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) & \mathbf{x} \in \mathbb{R}^n; \mathbf{u} \in \mathbb{R}^m \\ \mathbf{z}(t) = \mathbf{N}\mathbf{x}(t) & \mathbf{z} \in \mathbb{R}^q. \end{cases}$$

The state-feedback minimizing the LQ cost:

$$J = \int_0^\infty (\mathbf{z}^T(t)\mathbf{z}(t) + \rho \mathbf{u}^T(t)\mathbf{u}(t)) dt = \int_0^\infty (\mathbf{x}^T(t)\mathbf{N}^T\mathbf{N}\mathbf{x}(t) + \rho \mathbf{u}^T(t)\mathbf{u}(t)) dt$$

where ρ is positive scalar, reads:

$$\mathbf{u}(t) = -\mathbf{K}_c \mathbf{x}(t) \text{ with } \mathbf{K}_c = \frac{1}{\rho} \mathbf{B}^T \mathbf{P}_c \text{ and } \mathbf{P}_c = \text{Ric} \begin{bmatrix} \mathbf{A} & -\frac{1}{\rho} \mathbf{B}\mathbf{B}^T \\ -\mathbf{N}^T\mathbf{N} & -\mathbf{A}^T \end{bmatrix}.$$

LQ synthesis- asymptotic properties (cont.)

Then:

- $\lim_{\rho \rightarrow 0}(\mathbf{P}_c) = \mathbf{P}_0$ exists and $\lim_{\rho \rightarrow 0}(J) = \mathbf{x}^T(0)\mathbf{P}_0\mathbf{x}(0)$
- furthermore, if the following assumption are met:
 - * $\dim(\mathbf{z}) \leq \dim(\mathbf{u})$ ($q \leq m$),
 - * the transfer $\mathbf{G}_{zu}(s) = \mathbf{N}(s\mathbf{I}_n - \mathbf{A})^{-1}\mathbf{B}$ is minimum phase (without zeros in right half-plane),

(5)

then $\mathbf{P}_0 = 0$.

this last property highlights that, in the case of non-minimal phase system between u and z , it is not possible to reduce indefinitely the index performance (or to increase the performance) even if the control signal is not weighted in the index ($\rho \rightarrow 0$).

LQ synthesis- asymptotic properties (cont.)

When $\rho \rightarrow \infty$: (minimum energy design)

- the stable eigenvalues of $\mathbf{A}$ are invariant by the LQ state feedback. If the open-loop system is stable then $\mathbf{K}_c = 0$,
- the unstable eigenvalues of $\mathbf{A}$ are assigned to their stable symmetric (w.r.t. the imaginary axis) values $\rightarrow \mathbf{K}_c \neq 0$.

When $\rho \rightarrow 0$: (pure performance design)

let n_z be the number of transmission zeros in the transfer $\mathbf{G}_{zu}(s)$, then:

- n_z closed-loop eigenvalues are assigned to the “stable” zeros of $\mathbf{G}_{zu}(s)$ and to the stable symmetric values of the “unstable” zeros,
- the $n - n_z$ remaining closed-loop eigenvalues are driven far to stable asymptotes in the left-half plane,
- $\mathbf{P}_c = 0$ if $\mathbf{G}_{zu}(s)$ is minimum phase (no “unstable” zeros),

LQ synthesis: Root square locus

This is the hamiltonian matrix eigenvalue (root) locus when ρ varies from $+\infty$ (minimal energy design) to 0 (pure performance design). The stable part of this locus highlights the evolution of the closed-loop dynamic according to ρ .

$$\begin{aligned} \mathbf{H} &= \begin{bmatrix} \mathbf{A} & -\frac{1}{\rho}\mathbf{B}\mathbf{B}^T \\ -\mathbf{N}^T\mathbf{N} & -\mathbf{A}^T \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ -\mathbf{N}^T\mathbf{N} & -\mathbf{A}^T \end{bmatrix} - \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \frac{1}{\rho} \begin{bmatrix} \mathbf{0} & \mathbf{B}^T \end{bmatrix} \\ &= \mathcal{A} - \mathcal{B}k\mathcal{C} \end{aligned}$$

The *root square locus* is the root locus of the fictitious system:

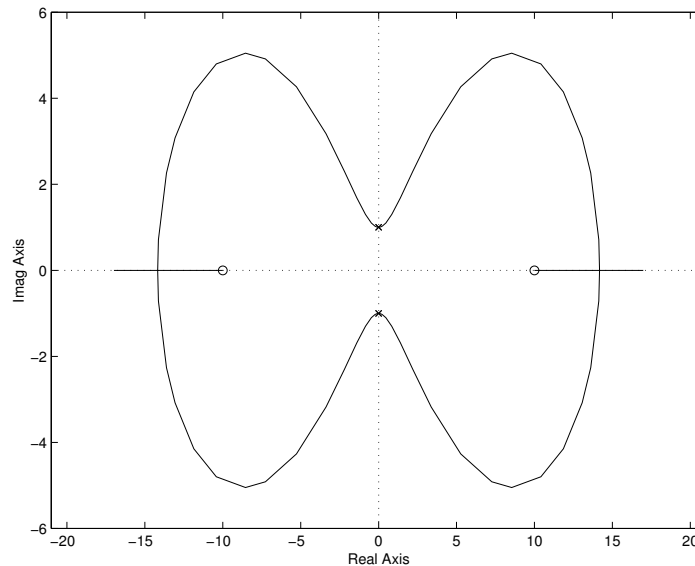
$$\mathcal{S} = \left[\begin{array}{c|c} \mathcal{A} & \mathcal{B} \\ \hline \mathcal{C} & 0 \end{array} \right] \quad \text{soit : } \mathcal{S}(s) = \mathbf{G}_{zu}^T(-s)\mathbf{G}_{zu}(s)$$

with a loop gain $k = 1/\rho$ varying from 0 to $+\infty$.

See function `rsqlocus.m` in toolbox `bib1` (available at: <http://personnel.isae.fr/daniel-alazard/matlab-packages/materials-for-jdmacs-2011.html>).

LQ synthesis : Root square locus example

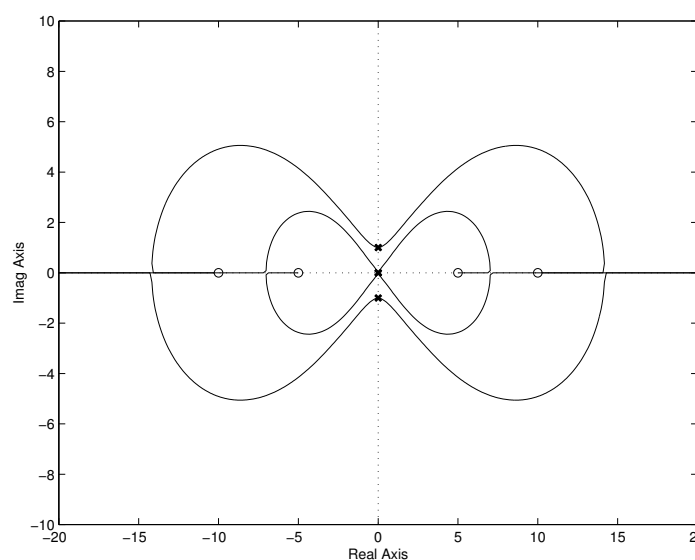
```
>> a=[0 1;-1 0];b=[0;1];n=[1 -0.1];
>> At=[a 0*a;-n'*n -a'];Bt=[b;0*b];Ct=[0*b' b'];
>> rlocus(At,Bt,Ct,0)
```



or just enter: `>> rsqlocus(a,b,n'*n,1).`

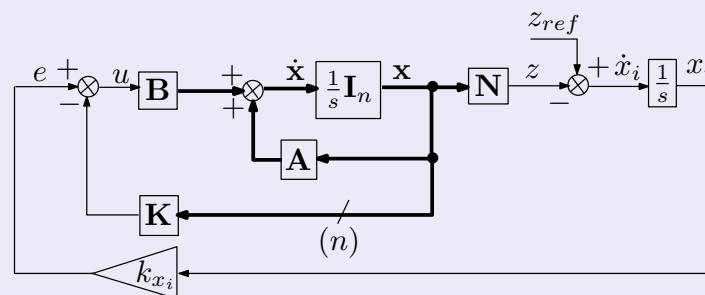
Root square locus - multivariable example

```
>> a=[0 1 0 0;-1 0 0 0;1 0 0 1;0 0 0 0];
>> b=[0 0;1 0;0 0;0 1];n=[1 -0.1 0 0;0 0 1 0.2];
>> rsqlocus(a,b,n'*n,eye(2,2));
```



Tuning of weights $\mathbf{Q}_x = \mathbf{N}^T \mathbf{Q} \mathbf{N}$ and $\mathbf{R}$

- $\mathbf{N}$ is given by the controlled outputs $\mathbf{z} = \mathbf{N}\mathbf{x}$ (specifications),
- the root square locus can be used to link the weights $\mathbf{Q}_x$ and $\mathbf{R}$ with closed-loop eigenvalues assignment,
- **Implicit reference model:** $J = \int_0^\infty (\dot{\mathbf{x}} - \mathbf{A}_d \mathbf{x})^T (\dot{\mathbf{x}} - \mathbf{A}_d \mathbf{x}) dt$ where $\mathbf{A}_d$ is the wanted closed-loop dynamics (with dynamic decoupling)
 $\Rightarrow \mathbf{Q}_x = (\mathbf{A} - \mathbf{A}_d)^T (\mathbf{A} - \mathbf{A}_d)$, $\mathbf{R} = \mathbf{B}^T \mathbf{B}$, $\mathbf{S} = (\mathbf{A} - \mathbf{A}_d)^T \mathbf{B}$,
- **Integral control:** $\mathbf{K}_a = [\mathbf{K} \quad k_{x_i}]$ is designed on the augmented model with: $\mathbf{x}_a = [\mathbf{x}^T, x_i]^T$ and $\dot{\mathbf{x}}_a = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ -\mathbf{N} & 0 \end{bmatrix} \mathbf{x}_a + \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix} u$.



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Norms for linear systems

H_2 norm: $\|\mathbf{G}(s)\|_2 = \left(\frac{1}{2\pi} \int_{-\infty}^{+\infty} \text{Trace} (\mathbf{G}^T(-j\omega)\mathbf{G}(j\omega))d\omega \right)^{1/2}.$

Interpretation: $\|\mathbf{G}(s)\|_2^2$ is the output signal energy when the input signal $\mathbf{u}(t)$ is an impulse (SISO case), that is $\mathbf{U}(s) = \mathbf{I}$, or $\|\mathbf{G}(s)\|_2^2$ is the variance of the output signal $\mathbf{y}(t)$ when $\mathbf{u}(t)$ is a random signal such that $\mathbf{U}(j\omega)\mathbf{U}^*(j\omega) = \mathbf{I}$ (uniform PSD, that is: a centered normalized white noise). $\|\mathbf{G}(s)\|_2$ is only definite for stable **strictly** proper ($\mathbf{D} = 0$) system (otherwise $\|\mathbf{G}(s)\|_2 = +\infty$).

H_∞ norm: $\|\mathbf{G}(s)\|_\infty = \sup_{\omega} \sigma_{\max} (\mathbf{G}(j\omega)) = \sup_{\mathbf{U}(s) \in H_2} \frac{\|\mathbf{Y}(s)\|_2}{\|\mathbf{U}(s)\|_2}.$

Interpretation: this norm is induced from the norm L_2 on signals. This norm is a measure of the largest magnitude of the system frequency-domain response $\mathbf{G}(j\omega)$.

State-space computation of the H_2 norm

Let $\mathbf{G}(s)$ be a stable strictly proper transfer: $\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$

$$\mathbf{G}(j\omega) \xrightarrow{\mathcal{F}} \mathbf{C}e^{\mathbf{A}t}\mathbf{B}$$

$$\frac{1}{2\pi} \int_{-\infty}^{+\infty} (\mathbf{G}^T(-j\omega)\mathbf{G}(j\omega))d\omega = \int_0^\infty \mathbf{B}^T e^{\mathbf{A}^T t} \mathbf{C}^T \mathbf{C} e^{\mathbf{A}t} \mathbf{B} dt \quad (\text{Parseval})$$

let us define $\mathbf{Q}_o = \int_0^\infty e^{\mathbf{A}^T t} \mathbf{C}^T \mathbf{C} e^{\mathbf{A}t} dt$ the **observability grammian**. $\mathbf{Q}_o$ is a definite positive (symmetric) matrix solution of the LYAPUNOV equation:

$$\begin{aligned} \mathbf{A}^T \mathbf{Q}_o + \mathbf{Q}_o \mathbf{A} + \mathbf{C}^T \mathbf{C} &= \mathbf{0} \\ \Rightarrow \|\mathbf{G}(s)\|_2^2 &= \text{trace} \mathbf{B}^T \mathbf{Q}_o \mathbf{B}. \end{aligned}$$

State-space computation of the H_∞ norm

Let $G(s)$ be a stable transfer: $\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$.

- $\|\mathbf{G}(s)\|_\infty$ cannot be computed analytically,
- The “peak” of $\sigma_{\max}(\mathbf{G}(j\omega))$ can be searched using a frequency gridding $\Rightarrow$ one can miss the “peak”!.
- Use the following property $\|\mathbf{G}(s)\|_\infty > \sigma_{\max}(\mathbf{D})$ and for all $\gamma > \sigma_{\max}(\mathbf{D})$, the 2 following assertions are equivalent:

(i) $\|\mathbf{G}\|_\infty < \gamma$,

(ii) the matrix

$$\mathbf{H}(\gamma) = \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & -\mathbf{A}^T \end{bmatrix} - \begin{bmatrix} \mathbf{0} & \mathbf{B} \\ \mathbf{C}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \gamma\mathbf{I} & \mathbf{D} \\ \mathbf{D}^T & \gamma\mathbf{I} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{C} & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}^T \end{bmatrix}$$

has no eigenvalue on the imaginary axis.

$\Rightarrow$ **Dichotomic search.**

State-space computation of the H_∞ norm

Proof in the case: $\mathbf{D} = 0$

$$\text{then: } \mathbf{H}(\gamma) = \begin{bmatrix} \mathbf{A} & \gamma^{-1}\mathbf{B}\mathbf{B}^T \\ -\gamma^{-1}\mathbf{C}^T\mathbf{C} & -\mathbf{A}^T \end{bmatrix}$$

Considering: $\phi(s) = \gamma^2\mathbf{I} - \mathbf{G}^T(-s)\mathbf{G}(s)$

$$\|\mathbf{G}\|_\infty < \gamma \Leftrightarrow \phi(j\omega) > 0 \quad \forall \omega \in \mathbb{R}$$

now $\phi(j\omega)$ is real and positive for $\omega \rightarrow \infty$, thus the sign of $\phi(j\omega)$ cannot change for $\omega \in [0, +\infty[$.

$\Rightarrow j\omega$ cannot be a zero of $\phi(s)$ or a pole of $\phi^{-1}(s)$.

Computing the dynamic state matrix $\mathbf{A}_{\phi^{-1}}$ of $\phi^{-1}(s)$, it can't be checked that $\mathbf{A}_{\phi^{-1}}$ and $\mathbf{H}(\gamma)$ are similar.

Exercise: compute H_2 and H_∞ norms of

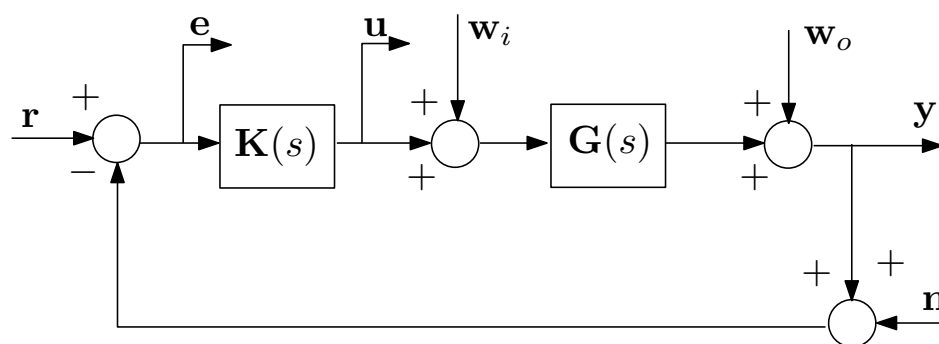
$$G(s) = \frac{\omega_0^2}{s^2 + 2\xi\omega_0 s + \omega_0^2}$$

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H_∞ requirements for servo-loop systems

Main transfers in a servo-loop system:



$$\begin{aligned}
 \mathbf{Y} &= \mathbf{GK}(\mathbf{I} + \mathbf{GK})^{-1}(\mathbf{R} - \mathbf{N}) + (\mathbf{I} + \mathbf{GK})^{-1}\mathbf{W}_o + \mathbf{G}(\mathbf{I} + \mathbf{KG})^{-1}\mathbf{W}_i \\
 \mathbf{E} &= (\mathbf{I} + \mathbf{GK})^{-1}(\mathbf{R} - \mathbf{W}_o - \mathbf{N}) - \mathbf{G}(\mathbf{I} + \mathbf{KG})^{-1}\mathbf{W}_i \\
 \mathbf{U} &= \mathbf{K}(\mathbf{I} + \mathbf{GK})^{-1}(\mathbf{R} - \mathbf{W}_o - \mathbf{N}) - \mathbf{KG}(\mathbf{I} + \mathbf{KG})^{-1}\mathbf{W}_i .
 \end{aligned}$$

- $\mathbf{S}(s) = (\mathbf{I} + \mathbf{G}(s)\mathbf{K}(s))^{-1}$: **Output sensitivity function**,
- $\mathbf{T}(s) = \mathbf{G}(s)\mathbf{K}(s)(\mathbf{I} + \mathbf{G}(s)\mathbf{K}(s))^{-1}$: **Complementary output sensitivity function**,
- $\mathbf{L}(s) = \mathbf{G}(s)\mathbf{K}(s)$: **Loop transfer function** (the loop is cut on $\mathbf{y}$).

H_∞ requirements for servo-loop systems

- Servo-loop performance: the transfer between the reference input r and the servo-loop error e must be “small”, mainly in low frequency (assuming e is a low-frequency signal):

$$\sigma_{\max}(\mathbf{S}(j\omega)) < |\mathbf{S}_{desired}(j\omega)| \quad \forall \omega$$

$$\Rightarrow \|\mathbf{W}_1 \mathbf{S}\|_\infty < 1 \quad \text{where} \quad \mathbf{W}_1 = \frac{1}{\mathbf{S}_{desired}} \quad \text{has a low-pass response.}$$

- Noise attenuation on the output: the transfer between n and y must be “small”, mainly in high frequency (assuming that n is a high frequency signal):

$$\Rightarrow \|\mathbf{W}_3 \mathbf{T}\|_\infty < 1 \quad \text{where} \quad \mathbf{W}_3 = \frac{1}{\mathbf{T}_{desired}} \quad \text{has a high-pass response.}$$

Frequency-domain management of the trade-off: $\mathbf{S} + \mathbf{T} = \mathbf{I}$

Control Design for aerospace vehicles

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H_∞ requirements for servo-loop systems

- high frequency attenuation of the control signal (actuator health): the transfer between the reference input r and the control signal u must be “small” in high frequency:

$$\Rightarrow \|\mathbf{W}_2 \mathbf{K} \mathbf{S}\|_\infty < 1 \quad \text{where} \quad \mathbf{W}_2 = \frac{1}{(\mathbf{K} \mathbf{S})_{desired}} \quad \text{has a high-pass response.}$$

- modulus margin: (reminder: let r be the radius of the circle centered at -1 , in the NYQUIST plane, and tangent to the frequency response of the loop transfer $L(s) = G(s)K(s)$)

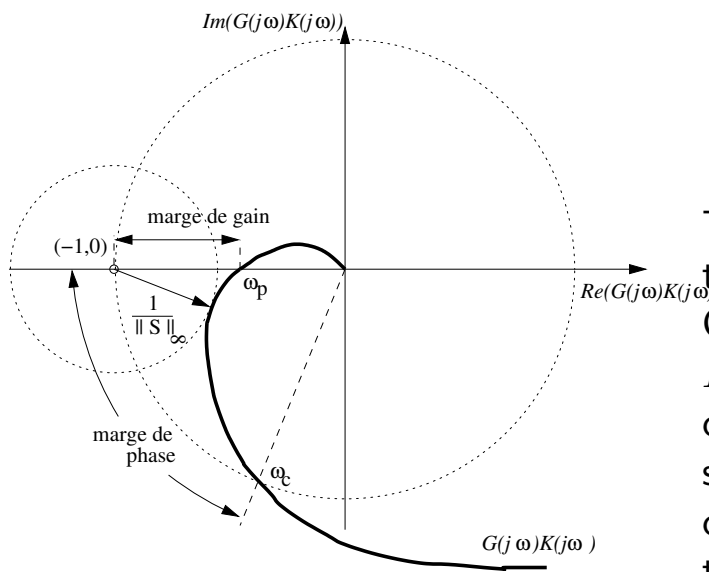
$$r = \min_{\omega} \sigma_{\min}(\mathbf{I} + \mathbf{L}(j\omega)) \quad \Leftrightarrow \quad \frac{1}{r} = \|\mathbf{S}\|_\infty$$

Increasing the modulus margin $\Leftrightarrow$ decreasing $\|\mathbf{S}\|_\infty$.

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Modulus margin



Gain, phase and modulus margins in the NYQUIST plane.

The modulus margin represents the smallest distance from the OL frequency-domain response $L(j\omega) = G(j\omega)K(j\omega)$ to the critical point and can be measured by the radius r of the circle centered on the critical point and tangent with $L(j\omega)$ response:

$$r = \min_{\omega} |1 + G(j\omega)K(j\omega)|.$$

Modulus margin

The modulus margin is the only stability margin which can be generalized to MIMO servo-loop systems, using singular values:

$$\text{MIMO case: } r = \min_{\omega} \sigma_{\min}(\mathbf{I} + \mathbf{G}(j\omega)\mathbf{K}(j\omega))$$

also:

$$r = \min_{\omega} \frac{1}{\sigma_{\max}(\mathbf{I} + \mathbf{G}(j\omega)\mathbf{K}(j\omega))^{-1}}$$

$$\frac{1}{r} = \max_{\omega} \sigma_{\max}(\mathbf{I} + \mathbf{G}(j\omega)\mathbf{K}(j\omega))^{-1} = \|\mathbf{S}(s)\|_{\infty}$$

Robustness to unstructured uncertainties

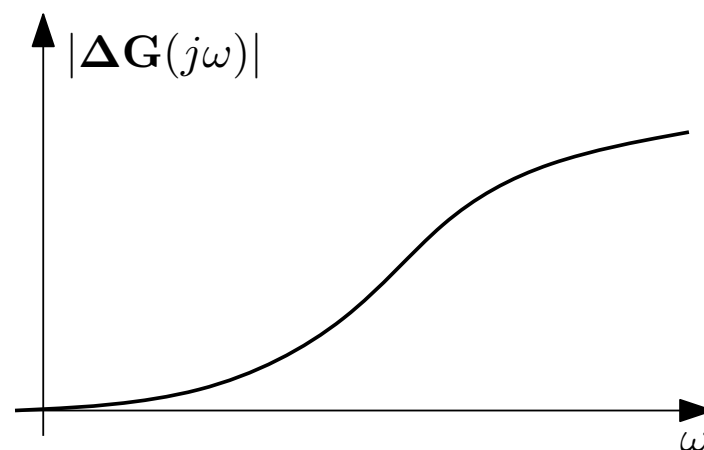
2 main kinds of uncertainties in the model:

- **parametric uncertainties** (ex: the mass of the aircraft during a long cruise flight changes because of fuel consumption): H_∞ design cannot take into account this kind of uncertainties (see μ -analysis et μ -synthesis chapters),
- frequency-domain uncertainties (or **unstructured uncertainties**, ex: neglected dynamic phenomena: actuators, sensors, $\dots$).

$$\mathbf{G}_{\text{true}}(s) = \mathbf{G}(s) + \Delta\mathbf{G}(s) \quad (\text{additive uncertainty}).$$

It is assumed that an upper bound of $\Delta\mathbf{G}(j\omega)$ exists for each frequency ω .

Unstructured uncertainties:



Typical profile for $|\Delta\mathbf{G}(j\omega)|$.

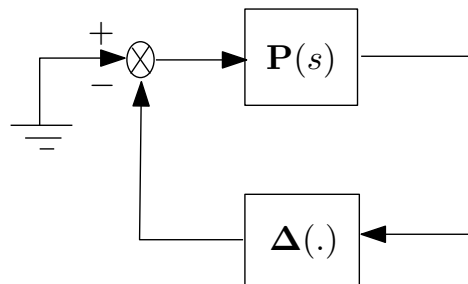
The uncertainty is upper-bounded by $W_a(s)|_{s=j\omega} \quad \forall \omega$:

$$\sigma_{\max}(\Delta\mathbf{G}(j\omega)) \leq |W_a(j\omega)| \quad \forall \omega$$

Small gain theorem

Considering the feedback loop depicted in the next Figure where the nominal system $\mathbf{P}(s)$ is linear, stationary and stable and the unstructured uncertainty $\Delta(\cdot)$ is an operator such that:

$$\sigma_{max}(\Delta(\omega)) \leq 1 \quad \forall \omega. \quad (6)$$



Small gain theorem - robust stability.

Small gain theorem

This loop is stable for all Δ constrained by (6) if and only if

$$\|\mathbf{P}(s)\|_{\infty} < 1,$$

This result works also when the uncertainty Δ is restricted to the space of (stationary) stable linear systems $\Delta(s)$ with a H_{∞} norm $\|\Delta(s)\|_{\infty} \leq 1$.
Practically: if the uncertainty is bounded by:

$$\sigma_{max}(\Delta(j\omega)) \leq |W_a(j\omega)| \quad \forall \omega$$

the small gain theorem is applied with $\|W_a(s)\mathbf{P}(s)\|_{\infty}$ and $\|\Delta(s)/W_a(s)\|_{\infty}$ (standardization of $\Delta(s)$).

Robust stability requires small gain in the frequency range where the uncertainty is big.

Robust stability conditions

Type of uncertainty	uncertainty upper bound	robust stability iff
$\mathbf{G} + \Delta$ (additive)	$\ \Delta(j\omega)\ _\infty \leq W_a(\omega) $	$\ W_a \mathbf{K}(\mathbf{I} + \mathbf{GK})^{-1}\ _\infty < 1$
$\mathbf{G}(\mathbf{I} + \Delta)$ (input mult.)	$\ \Delta(j\omega)\ _\infty \leq W_{mi}(\omega) $	$\ W_{mi} \mathbf{KG}(\mathbf{I} + \mathbf{KG})^{-1}\ _\infty < 1$
$(\mathbf{I} + \Delta)\mathbf{G}$ (output mult.)	$\ \Delta(j\omega)\ _\infty \leq W_{mo}(\omega) $	$\ W_{mo} \mathbf{GK}(\mathbf{I} + \mathbf{GK})^{-1}\ _\infty < 1$
$\mathbf{K} + \Delta$ (feedback)	$\ \Delta(j\omega)\ _\infty \leq W_k(\omega) $	$\ W_k \mathbf{G}(\mathbf{I} + \mathbf{KG})^{-1}\ _\infty < 1$

Outline

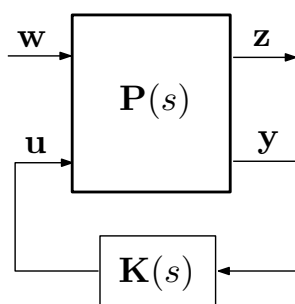
- 1 State-feedback for eigen-structure assignment
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The H_∞ standard problem

Standard problem: an augmented model embedding the model $\mathbf{G}(s)$ (transfer from $\mathbf{u}$ to $\mathbf{y}$) and frequency-domain specifications (exogenous input $\mathbf{w}$ and controlled variables $\mathbf{z}$).



Standard LFT: $F_l(\mathbf{P}, \mathbf{K})$

Representations:

$$\bullet \mathbf{P}(s) = \begin{bmatrix} \mathbf{P}_{zw}(s) & \mathbf{P}_{zu}(s) \\ \mathbf{P}_{uw}(s) & \mathbf{P}_{yu}(s) \end{bmatrix}$$

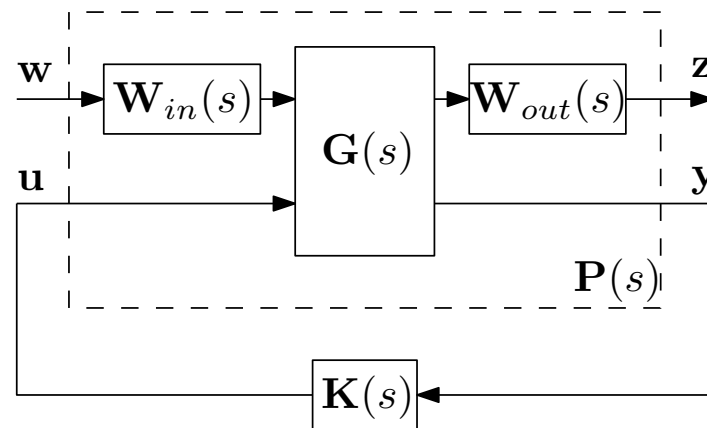
$$\bullet \begin{bmatrix} \dot{\mathbf{x}} \\ \mathbf{z} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B}_1 & \mathbf{B}_2 \\ \mathbf{C}_1 & \mathbf{D}_{11} & \mathbf{D}_{12} \\ \mathbf{C}_2 & \mathbf{D}_{21} & \mathbf{D}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{w} \\ u \end{bmatrix}$$

H_∞ control problem: find a stabilizing controller $\mathbf{K}(s)$ such that:

$$\|F_l(\mathbf{P}(s), \mathbf{K}(s))\|_\infty \leq \gamma \quad \text{for a given scalar } \gamma$$

with: $\|\mathbf{F}(s)\|_\infty = \max_\omega \bar{\sigma}(\mathbf{F}(j\omega))$ where $\bar{\sigma}(\mathbf{M})$ is the highest singular value of matrix $\mathbf{M}$.

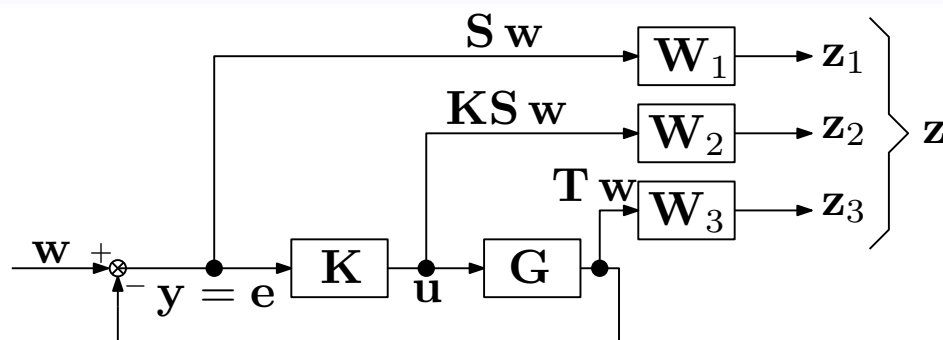
Standard problem with frequency-domain weights



General standard problem with frequency-domain weights on exogenous inputs and controlled outputs.

- poles of $W_{in}(s)$ are uncontrollable from u ,
- poles of $W_{out}(s)$ are unobservable by y .

Ex: the mixed-sensitivity problem

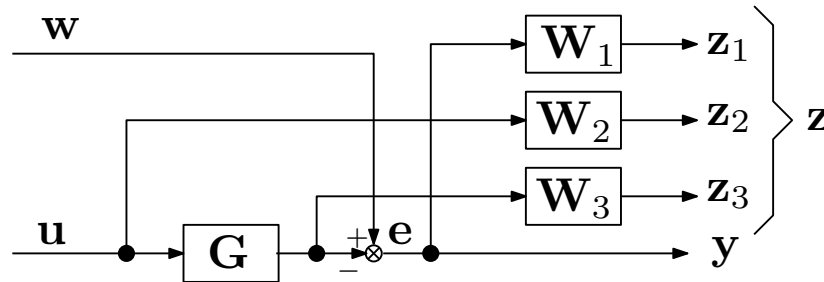


- $T_{w \rightarrow z_1} = W_1 S$ with $S = (I + GK)^{-1}$: servo-loop performance,
- $T_{w \rightarrow z_3} = W_3 T$ with $T = GK(I + GK)^{-1}$: measurement-noise attenuation,
- $T_{w \rightarrow z_2} = W_2 KS$: actuator health.

Frequency-domain management of the trade-off $S + T = I$:

$\Rightarrow W_1$ is low-pass, W_2 and W_3 are high-pass.

Ex: the mixed-sensitivity problem



Standard problem $P(s)$ for mixed-sensitivity.

More generally, from a SIMULINK file describing the block-diagram of $P(s)$ and the function `linmod`, a state-space realization of $P(s)$ (matrices A , B_1 , B_2 , C_1 , C_2 , D_{11} , D_{12} , D_{21} , D_{22}) can be computed. Warning: don't forget to:

- plug y to the last output ports,
- plug u to the last input ports.

Mixed-sensitivity H_∞ standard problem

Find a controller $K(s)$ stabilizing the closed-loop model and such that:

$$\left\| \begin{bmatrix} W_1 S \\ W_2 KS \\ W_3 T \end{bmatrix} \right\|_\infty < 1.$$

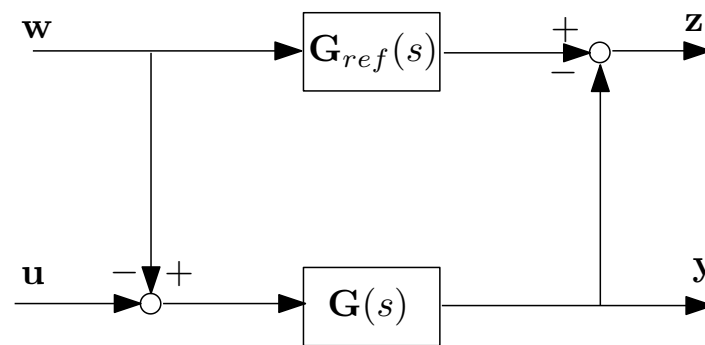
Remark:

$$\left\| \begin{bmatrix} W_1 S \\ W_2 KS \\ W_3 T \end{bmatrix} \right\|_\infty < 1 \text{ (a)} \Rightarrow \begin{cases} \|W_1 S\|_\infty < 1 \\ \|W_2 KS\|_\infty < 1 \\ \|W_3 T\|_\infty < 1 \end{cases} \text{ (b)} . \quad (7)$$

(7.a) is a sufficient but not necessary condition for (7.b).

Other standard problems

Example: Reference model $\mathbf{G}_{ref}(s)$ following



Warning: the ideal controller $\widehat{\mathbf{K}}(s)$ is such that:

$$F_l(\mathbf{P}(s), \widehat{\mathbf{K}}(s)) = \mathbf{G}_{ref}(s) - (\mathbf{I} + \mathbf{G}(s)\widehat{\mathbf{K}}(s))^{-1}\mathbf{G}(s) = \mathbf{0} .$$

Thus, the ideal optimal performance index value $\hat{\gamma}_{ideal}$ is 0 (and not 1). This approach is quite different from the mixed-sensitivity approach where the gap between $\hat{\gamma}$ and 1 gives the distance to the frequency-domain templates.

H_∞ problem solvers

General pb: given $\mathbf{P}(s)$, find a stabilizing controller $\widehat{\mathbf{K}}(s)$ such that:

$$\widehat{\mathbf{K}}(s) = \arg \min_{\mathbf{K}(s)} (\|F_l(\mathbf{P}(s), \mathbf{K}(s))\|_\infty) .$$

Full order design: it is possible to parameterize the searched controller (decision variable) in order to convexify the optimization problem and to find the global optimum ($\hat{\gamma} = \min_{\mathbf{K}(s)} \|F_l(\mathbf{P}(s), \mathbf{K}(s))\|_\infty$). But the order of resulting controller is equal to the order of the standard problem (and could raise implementation problems).

$\Rightarrow$ RICCATI based solver (DGKF algorithm presented below) or LMI-based solver.

Fixed-structure controller design (for instance, reduced order): if the structure of the controller is *a priori* fixed, then non-smooth optimization algorithms provide a local optimum (help systune, see also: hifoo toolbox). They can also relax some restrictive assumptions required by full order solvers.

Multi-objective synthesis

Multi-channel H_∞ problem: Fixed-structured H_∞ synthesis can also cope with a structured performance index (that is: the structure of the standard problem).

Example: 2 channel control problem:

$$\begin{aligned} & \min_{\mathbf{K}(s)} \left\| \begin{array}{cc} F_l(\mathbf{P}, \mathbf{K})(1, 1) & 0 \\ 0 & F_l(\mathbf{P}, \mathbf{K})(2, 2) \end{array} \right\|_\infty \\ &= \min_{\mathbf{K}(s)} \max\{\|F_l(\mathbf{P}, \mathbf{K})(1, 1)\|_\infty, \|F_l(\mathbf{P}, \mathbf{K})(2, 2)\|_\infty\} \end{aligned}$$

Mixed H_2/H_∞ control problem: A lot of control problems can be expressed as the minimization of an H_2 performance index under an H_∞ constraint (see `hinfmix` for the full order design case or `hifoo` and `systune` for fixed-structure design case).

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Ex: the 2-objectives S – KS H_∞ problem

Let us consider the second order model of an unstable aerospace vehicle:

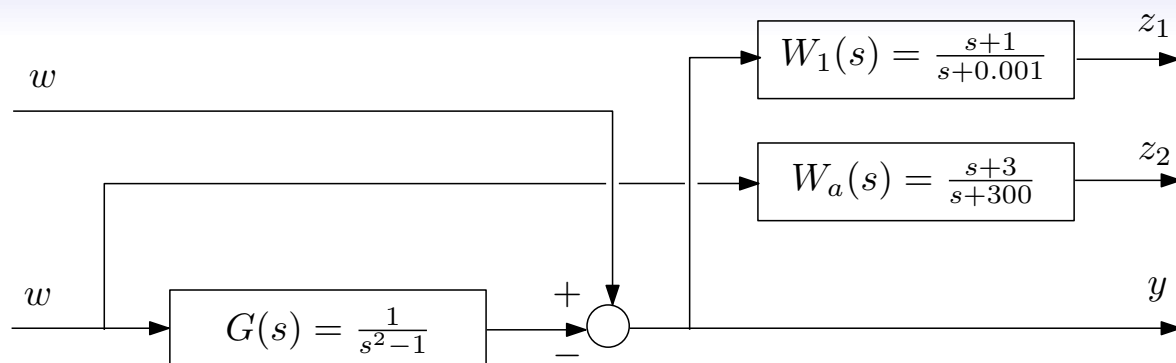
$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{s^2 - 1}$$

The objective is to design a **stabilizing** H_∞ feedback controller $K(s)$ such that:

- $\gamma_1 = \|W_1(s)S(K(s))\|_\infty \leq 1$ with $W_1(s) = \frac{s+1}{s+0.001}$. $W_1(s)$ is in fact the regularized weight of $\frac{s+1}{s}$ chosen to impose an integral effect and a 1 rd/s bandwidth,
- robust stability to unstructured additive uncertainty
 $\gamma_2 = \|W_a(s)K(s)S(K(s))\|_\infty \leq 1$ with $W_a(s) = \frac{s+3}{s+300}$. $W_a(s)$ is in fact the regularized filter of $0.01(1 + s/3)$ (the upper bound of additive uncertainty).

$\Rightarrow$ a 2 objectives control problem.

The standard H_∞ problem $P(s)$ such that $F_l(P, K) = \begin{bmatrix} W_1 S \\ W_1 K S \end{bmatrix}$



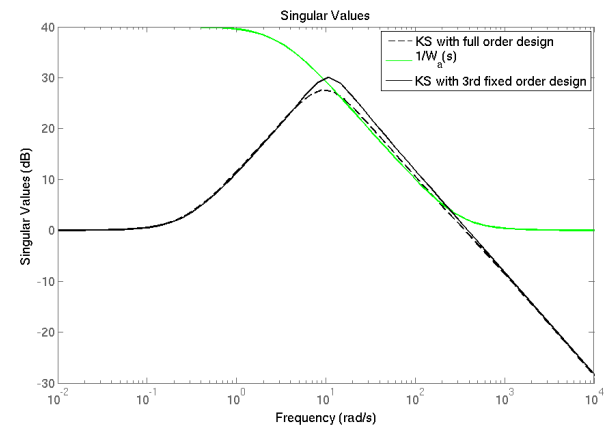
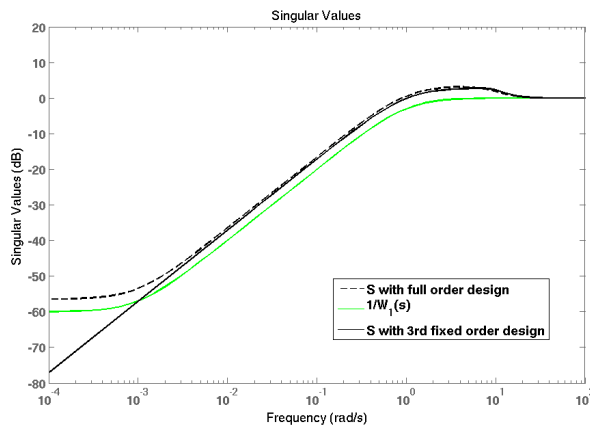
- Full order H_∞ design (hinfsyn) can only cope with:
 $\min_K \|F_l(\mathbf{P}, K)\|_\infty$. Indeed, a sufficient condition reads:

$$\|F_l(\mathbf{P}, K)\|_\infty \leq 1 \Rightarrow \gamma_1 \leq 1 \text{ and } \gamma_2 \leq 1.$$

- Fixed-structure H_∞ design (hinfstruct) can cope with:
 $\min_K \max_K(\gamma_1, \gamma_2)$ and can work with a pure integral third order controller: $K(s) = \frac{b_2 s^2 + b_1 s + b_0}{s(s^2 + a_1 s + a_0)}$.

Results

	full order	fixed structure
$\ F_l(\mathbf{P}, K)\ _\infty$	1.4974	1.7763
γ_1	1.4973	1.3919
γ_2	1.0093	1.3814
$\max(\gamma_1, \gamma_2)$	1.4973	1.3919



Conclusions

The H_∞ design of a suitably structured controller allows to circumvent most of methodological problems encountered with full order design:

- avoiding the plant model local inversion in an over-parameterized controller,
- simplifying the set-up of the H_∞ standard problem,
- short-cutting controller reduction when the design concerns high-order model (for instance flexible structures),
- simplifying the controller on-board implementation

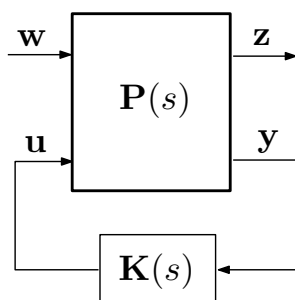
Fixed-structure H_∞ synthesis can also be used to optimize high level system parameters: $\Rightarrow$ towards system/controller co-design.

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The H_2 standard problem

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Standard LFT: $F_l(\mathbf{P}, \mathbf{K})$

Representations:

$$\bullet P(s) = \begin{bmatrix} \mathbf{P}_{zw}(s) & \mathbf{P}_{zu}(s) \\ \mathbf{P}_{uw}(s) & \mathbf{P}_{yu}(s) \end{bmatrix}$$

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H_2 control problem: find a stabilizing controller $\hat{\mathbf{K}}(s)$ such that:

$$\hat{\mathbf{K}}(s) = \arg \min_{\mathbf{K}(s)} \|F_l(\mathbf{P}(s), \mathbf{K}(s))\|_2$$

Control of aerospace vehicles
Labwork on Matlab/Simulink - Modal, LQ and H_∞
control design

Lateral flight control design

The A/C lateral flight model G is given by the state-space representation:

$$\begin{bmatrix} \dot{\beta} \\ \dot{p} \\ \dot{r} \\ \dot{\phi} \\ n_y \\ p \\ r \\ \phi \end{bmatrix} = \begin{bmatrix} -0.140 & 0.053 & -0.999 & 0.047 & 0 & 0.030 \\ -2.461 & -0.992 & 0.262 & 0 & 0.404 & 0.260 \\ 1.585 & -0.041 & -0.267 & 0 & 0. & -0.680 \\ 0 & 1.000 & 0.053 & 0 & 0 & 0 \\ 0.0433 & -0.0003 & 0.0016 & 0 & 0.0001 & -0.0075 \\ 0 & 1.0000 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.0000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.0000 & 0 & 0 \end{bmatrix} \begin{bmatrix} \beta \\ p \\ r \\ \phi \\ dp \\ dr \end{bmatrix}. \quad (1)$$

- 4 states: β (sideslip angle, rd), p (roll rate, rd/s), r (yaw rate, rd/s), ϕ (roll angle, rd),
- 2 control signals: dp (aileron angle rd) and dr (rudder angle, rd),
- 4 measurements: n_y (lateral acceleration, g), p (roll rate), R (yaw rate), ϕ (roll angle)

1 Static feedback and feedforward design

The lateral flight control law is depicted on the following Figure where K is static output feedback and H is a static (2×2) feedforward designed to met the steady state specification:

$$\lim_{t \rightarrow \infty} \begin{bmatrix} \beta(t) \\ \phi(t) \end{bmatrix} = \begin{bmatrix} \beta_{ref} \\ \phi_{ref} \end{bmatrix}$$

For each design (modal and LQ):

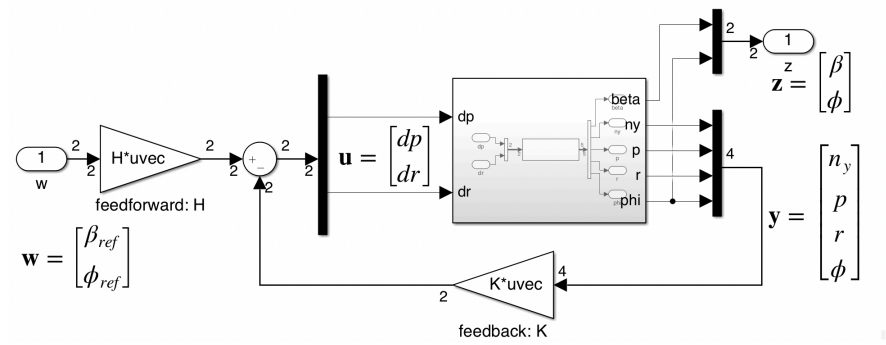


Figure 1: Lateral flight control law.

- the state feedback ($u = -K_x x$) will be first designed,
- the number of measurements (4) being equal to the number of states, the output feedback K equivalent to the state feedback K_x will be computed,
- the modulus margin will be analyzed,
- the responses of β and ϕ to initial conditions on β and ϕ will be simulated and analyzed,
- finally, the feed-forward H will be computed and the step responses of the closed loop system (Figure 1) will be analysed.

1.1 Eigen-structure assignment

The requirements:

- the roll mode must be assigned to -1.1 and decoupled from β ,
 - the spiral mode must be assigned to -1 and decoupled from β ,
 - the Dutch roll mode must be assigned to $-1 \pm 1.3i$ and decoupled from ϕ .
- compute and analyze the control law assigning only the 4 eigenvalues (function `place`),
- compute and analyze the control law which meets also decoupling requirements.

1.2 LQ control

The state feedback K_x is now computed to minimize the performance index:

$$J = \int_0^\infty (q_r r^2 + q_\phi \phi^2 + dr^2 + dp^2) dt$$

- using q_r and q_ϕ (try and error procedure), compute a state feedback K_x such that the Dutch roll damping ratio is at least 0.6 and the spiral mode is faster than 1 (rd/s). Analyze such a solution and comment decoupling properties,

Implicit reference model: the performance index is changed to ($x = [\beta, p, r, \phi]^T$):

$$J = \int_0^\infty (\dot{x} - A_d x)^T (\dot{x} - A_d x) dt$$

with:

$$A_d = \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & -2\xi_r\omega_r & 0 & -\omega_r^2 \\ \omega_l^2 & 0 & -2\xi_l\omega_l & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \quad \text{et : } \omega_r = 1.049 \text{ rd/s}, \omega_l = 1.64 \text{ rd/s}, \xi_r = 1.001, \xi_l = 0.61 .$$

Justify such a performance index. Compute the weighting matrices Q , R and S of the equivalent LQ performance index. Compute and analyse the corresponding solution.

2 H_∞ design of a 2 d.o.f control law

The H_∞ design of a 2 degrees-of-freedom control law is now considered:

$$\begin{bmatrix} dp \\ dr \end{bmatrix} = C_{2\text{dof}}(s) \begin{bmatrix} \beta_{ref} \\ \phi_{ref} \\ n_y \\ p \\ r \\ \phi \end{bmatrix} .$$

The proposed standard problem is depicted in the following Figure. 2.

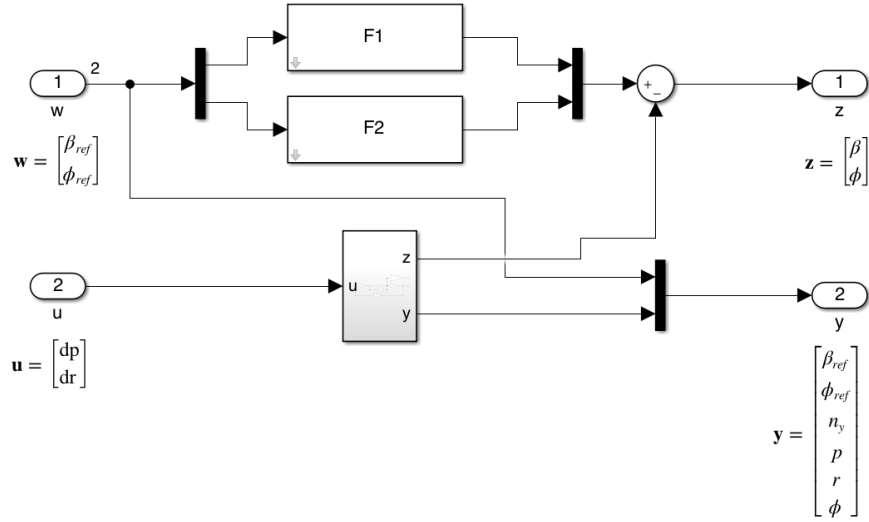


Figure 2: Standard problem for reference model following.

- comment such a standard problem,
- choose the weighting filters $F1(s)$ and $F2(s)$ to obtain a correct closed-loop behavior (decoupling, settling times),
- design an H_∞ full order controller (function `hinfsyn`) and a static controller (function `systune`). Analyze both solutions: step responses to reference inputs ($\beta_{consigne}$ and $\phi_{consigne}$) and the responses to an initial condition on β (due to a lateral wind gust),
- (-) design a second order controller minimizing the H_2 norm of closed transfer from w to u such that the H_∞ norm of the closed loop transfer from w to z is lower than 0.05.